UNIT → 4

$\exists \to$ There exists

Date ....

UNIT → 4

→ Vector Space

A vector space V over the field F is a non empty set together with 2 operations called addition and scalar multiplication such that the following axioms holds:-

- If $u, v \in V$ then $u+v \in V$
- If $a \in F$ and $u \in V$ then $a \cdot u \in V$ [Scalar multip.
- $u + (v+\omega) = (u+v)+\omega \quad \forall\ u, v, \omega \in V$ [ASSO.
- $u+v = v+u \quad \forall\ u, v \in V$ [commutative]
- $\exists$ element 0 such that $u + 0 = 0 + u = u \quad \forall\ u \in V$
- $\forall\ u \in V \quad \exists\ (-u) \in V$ such that $u + (-u) = 0$
- For ~~any~~ all $a, b \in F$ and $u, v \in V$
  - (I) $a \cdot (u+v) = a \cdot u + a \cdot v$
  - (II) $(a+b) u = au + bu$
  - (III) $(a \cdot b) u = a(bu)$
  - (IV) $1 \cdot u = u$

Identity element

Q Prove that set of all vectors in a plane ~~over~~ R over the feild of real no is a vector space w.r.t vector add & scalar multiplication

$$V = R^2 \qquad F = R$$

$$V = \{(x, y) : x, y \in R\}$$

Suppose $u = (x_1, y_1) \quad v = (x_2, y_2)$

then $u + v = (x_1 + x_2, y_1 + y_2) \in V$

Property 1 holds

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Date ....

suppose $a \in R$, $(x_1, y_1) \in V$
$a \cdot (x_1, y_1) = (ax_1, ay_1) \in V$
Property -2 Holds

suppose $(x_1 y_1)\ (x_2 y_2)\ (x_3 y_3) \in V$
$(x_1, y_1) + ((x_2, y_2) + (x_3\ y_3)$
$[(x_1 y_1) + (x_2\ y_2)] + (x_3 y_3)]$
P-3 holds

suppose $(x_1 y_1)$, $(x_2\ y_2) \in V$
$(x_1 y_1) + (x_2 y_2) = (x_2 y_2) + (x_1 y_1)$
P-4 holds

suppose $(x_1, y_1) \in V$
$(0, 0) \rightarrow$ zero element
$V + 0 = (x_1 + 0, y_1 + 0)$
$= (x_1, y_1)$
P - 5 holds

suppose $(x_1, y_1) \in V$
$V + x = 0$
$(x_1, y_1) + x = 0$
$x = (-x_1, -y_1) \in V$
P-6 holds

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Date ....

Q Prove that C(R) is a vector space but R(C) isn't a vector space

C(R)

Let us consider $a \in R$ & $z \in C$

$\Rightarrow a \cdot z \in C$

Real × complex ε complex

||ly check others, they satisfy ⇒ It's vector space

R(C)

Let us consider $z \in C$ & $b \in R$

$b \cdot c \in C$

Real × complex = complex

Hence it's not real $\notin R$

So it's not a vector space

→ Sub - space

A non-empty subset W of a vector space V(F) is said to be a sub space of V if

- $au + bv \in W \quad \forall a, b \in F$ & $u, v \in W$
- $W \subseteq V$

Q Let R be the feild of real no's. Then prove or disprove that $W = \{(x, 2y, 3z) : x, y, z \in R\}$ be a sub space of $R^3(R)$

Let $a, b \in R$, $u = (x_1, 2y_1, 3z_1)$, $v = (x_2, 2y_2, 3z_2)$ $\in W$

$au + bv$

$(ax_1 + bx_2, 2ay_1 + 2by_2 + 3az_1 + 3bz_2)$

$ax_1 + bx_2 \in R$

$ay_1 + by_2 \in R$

$az_1 + bz_2 \in R$

Hence, s

$(ax_1 + bx_2, 2(ay_1 + by_2), 3(az_1 + bz_2)) \in W$

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Date ....

Q show that $W = \{(x, +1, y, z) : x, y, z \in R\}$ is~~nt~~ a subspace of $R^3(R)$

$a, b \in R$ $\quad (x_1 + 1 ; y_1, z_1) \in W$
$\quad (x_2 + 1, y_2, z_2)$

$(ax_1 + a + bx_2 + b + 1 - 1 ; ay_1 + by_2 ; az_1 + bz_2)$

~~Let~~ $\cancel{a = 2 \quad b = 3}$

$\cancel{(2x_1 + 3x_2 + 5 ; 2y_1 + 3y_2 ; 2z_1 + 3z_2)}$

~~Let~~ $\cancel{ax_1 + bx_2 + 4 = K \in R}$

Let $a + b - 1 = K \in R$

$(ax_1 + bx_2 + K + 1 ; ay_1 + by_2, az_1 + bz_2)$

$ax_1 + bx_2 + K \in R \quad \because x_1, x_2, K \in R$
$ay_1 + by_2 \in R$
$az_1 + bz_2 \in R$

Hence, it's $\in W$
So it's sub space

Date ....

Q $W = \{(x, y, z) \mid x + y + z = 1, \; x, y, z \in R\}$
$R^3(R)$

$\cancel{ax + by}$ $\quad \cancel{a = 1} \; \cancel{(1, -1, 0)}$
$\cancel{b = 0} \; 2 \; \cancel{(-1, 0, 1)}$
α

$\cancel{1 + 1, -1, -1}$
$\cancel{-1, -1, 2}$

$u = (1, 0, 0) \quad v = (0, 1, 0)$
$a = 1 \qquad b = 2$
$au + bv = (1, 0, 0) + (0, 2, 0)$
$= (1, 2, 0)$
Here, $1 + 2 + 0 = 3 \neq 1$
Hence, $(1, 2, 0) \notin W$
It's not a sub space

Q Prove or disprove $W = \{(x, y, z), \; x, y, z \in N\}$ subspace of $R^3(R)$ or not

$a = \sqrt{3} \qquad b = 1$
$u = (1, 2, 3) \qquad v = (2, 3, 4)$

$(\sqrt{3} + 2, \; 2\sqrt{3} + 3, \; 3\sqrt{3} + 4) \notin W$

so it's not a sub space.

Date ....

Q $W = \left\{ \begin{bmatrix} x & y \\ z & 0 \end{bmatrix} : x, y, z \in \mathbb{R} \right\}$ Let V be the vector space of all square matrices over R

$$\begin{bmatrix} ax_1 + bx_2 & ay_1 + by_2 \\ az_1 + bz_2 & 0 \end{bmatrix} \in W$$

so, it's a sub space

Date ....

→ Span (S)

Let S be a collection of vectors then span(S) is the collection of all the possible linear combos of the vectors of S

Q Find span(S) for $S = \{(1,0,0), (0,1,0)\}$ where $S \subseteq R^3(R)$

$$\text{span}(S) = \{a(1,0,0) + b(0,1,0) : a, b \in R\}$$

$$= \{(a, b, 0) : a, b \in R\}$$

Q Find span(S) where $S = \{(2,3,1), (2,0,0)\}$ over R

$$\text{span}(S) = \{((2a+2b), 3a, a) : a, b \in R\}$$

→ Basis of a vector space

A subset B of a vector space V(F) is said to be basis of the vector space iff :-

1. All the vectors in B are linearly independent
2. span(B) = V(F)

Q Check which of these is a basis for $R^2(R)$

i) $B = \{(1,0), (0,1)\}$ ii) $B = \{(1,0), (1,1)\}$

iii) $B = \{(1,0), (2,0)\}$ iv) $B = \{(1,2), (3,6)\}$

i) $v_1 = (1,0)$ $v_2 = (0,1)$

$$c_1 v_1 + c_2 v_2 = (0,0)$$

$$(c_1, c_2) = (0,0)$$

$$c_1 = c_2 = 0$$

Linearly Ind. ✓

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$\text{span}(B) = \{(a, b) : a, b \in R\}$ ✓

Basis ✓

II) $(c_1 + c_2, \ c_2) = (0, 0)$

$c_1 + c_2 = 0 \qquad c_2 = 0$

$c_1 = 0$

L·I·D ✓

$\text{span}(B) = \{(a+b, \ b) : a, b \in R\}$

Basis ✓

$\Rightarrow$ Basis for $R^2(R)$ is not unique

III) $(c_1 + 2c_2, \ 0) = (0, 0)$

$c_1 = -2c_2$

LID ✗

Basis ✗

IV) $(c_1 + 3c_2, \ 2c_1 + 8c_2) = (0, 0)$

$c_1 = -3c_2$

LID ✗

Basis ✗

→ Dimension of a vector space

The number of elements present in the basis of a vector space is called dimension of a vector space. (It's unique)

Date ....

Q Find the dimension of vector space $R^3(R)$

$\cancel{B = \{(x, y, z) : x, y, z \in R\}}$

$\cancel{c_1 x_1 + c_2 x_2 = 0}$
$\cancel{c_1 y_1 + c_2 y_2 = 0}$
$\cancel{c_1 z_1 + c_2 z_2 = 0}$

$B = \{(1,0,0), (0,1,0), (0,0,1)\}$

$c_1 = 0$
$c_2 = 0$
$c_3 = 0$ LID ✓

$\text{span}(B) = \{(a,b,c) : a, b, c \in R\} \subseteq R^3(R)$

So, it's basis

Dimension of $R^3(R) = 3$

Date ....

→ Linear Transformation

A linear transformation is a function between 2 vector spaces that preserves vector addition and scalar multiplication

Let U & V be 2 vector spaces over same field F then a map $T : U \longrightarrow V$ is called a linear transformation if

i) $T(u+v) = T(u) + T(v)$

ii) $T(au) = aT(u)$

Q Which of these are linear transformation

(i) $T : R^3 \rightarrow R^3$ defined by

$$T(x, y, z) = (y, -x, -z)$$

(ii) $T : R^3 \rightarrow R^2$ defined by

$$T(x, y, z) = (2x - 3y, 7y + 2z)$$

1) $u = (x_1, y_1, z_1)$ $\quad v = (x_2, y_2, z_2)$

LHS

$u + v = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$

$T(u+v) = (y_1 + y_2, -x_1 - x_2, -z_1, -z_2)$

RHS

$T(u) = (y_1, -x_1, -z_1)$

$T(v) = (y_2, -x_2, -z_2)$

$T(u) + T(v) = (y_1 + y_2, -x_1 - x_2, -z_1 - z_2)$

LHS = RHS

Let $a \in F$, LHS:-

$au = (ax_1, ay_1, az_1)$

$T(au) = (ay_1, -ax_1, -az_1)$

RHS

$aT(u) = a[(y_1, -x_1, -z_1)]$

$= (ay_1, -ax_1, -az_1)$

LHS = RHS

So it's L.T.

II) $u = (x_1 y_1 z_1)$ $v = (x_2 y_2 z_2)$

LHS

$u+v = (x_1+x_2, y_1+y_2, z_1+z_2)$

$T(u+v) = (2x_1+2x_2-3y_1-3y_2, 7y_1+7y_2+2z_1+2z_2)$

RHS

$T(u) = (2x_1-3y_1, 7y_1+2z_1)$

$T(v) = (2x_2-3y_2, 7y_2+2z_2)$

$T(u)+T(v) = (2x_1+2x_2-3y_1-3y_2, 7y_1+7y_2+2z_1+2z_2$

Let $a \in F$ LHS :-

$au = (ax_1, ay_1, az_1)$

$T(au) = (2ax_1-3ay_1, 7ay_1+2az_1)$

Date ....

RHS

$T(u) = (2x_1 - 3y_1 \quad , 7y_1 + 2z_1)$

$aT(u) = (2ax_1 - 3ay_1, \; 7ay_1 + 2az_1)$

LHS = RHS

Property ① & ② satisfies
So it's L.T.

iii) $T : R^2 \to R^2$ defined by
$T(x, y) = (x^2, y)$

$u = (x, y_1) \qquad v = (x_2 \; y_2)$
$u + v = (x_1 + x_2, \quad y_1 + y_2)$

$T(u+v) = (x_1^2 + x_2^2 + 2x_1 x_2 \quad , y_1 + y_2)$

$T(u) = (x_1^2, y_1)$
$T(v) = (x_2^2, y_2)$

$T(u) + T(v) = (x_1^2 + x_2^2 \quad , y_1 + y_2)$

LHS $\neq$ RHS

Not a L.T.

Date ....

→ Null space / Kernel of a linear transformation
Let $T: U \to V$ be a linear transformation. The null space / Kernel of T is the sub set of U consisting of all the vectors whose image under T is 0 (vector) and is denoted by $Ker(T)$ or $N(T)$

$$Ker(T) = \{ u \in U : T(u) = 0 \}$$

NOTE : The dimension of $ker(T)$ is known as nullity

→ Range of Linear Transformation / Image set of linear transformation
Let $T: U \to V$ be a linear transformation then range of T is the set of all vectors that are image under T of vectors in U
It's denoted by $Range(T)$ or $R(T)$

$$R(T) = \{ T(u) \in V : u \in U \}$$

NOTE
Dimension of range set is known as Rank(T)

Q Find the $ker(T)$ & $Range(T)$ where $T \to R^3 \to R^2$ such that $T(x, y, z) = (x+y, y+z)$

Let $u = (x_1, y_1, z_1)$

$T(u) = (x_1 + y_1, y_1 + z_1) = \vec{0}$

$(x_1 + y_1, y_1 + z_1) = (0, 0)$

$x_1 = -y_1$ & $y_1 = -z_1$

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$Ker(T) = \{(x, -x, x) : x \in R\}$

Q Let $T: R^2 \to R^2$

$T(x, y) = (-x, 0)$

$u = (x_1, y_1)$

$T(u) = T(x_1, y_1) = 0$

$\Rightarrow (-x_1, 0) = (0, 0)$

$\Rightarrow x_1 = 0$

$Ker(T) = \{(0, y) : y \in R\}$

Range $=$ span $\{(1, 0)\}$

$R = \{(-x, 0), x \in R\}$

Q $T: R^2 \to R$

$T(x, y) = (x+y, x-y)$

$x = -y$ & $x = y$

$\Rightarrow x = y = 0$

$Ker(T) = \{(0, 0)\}$

Range $= \{(x+y, x-y) : x, y \in R\}$

or

$R =$ span $\{(1,0), (0,1)\} = R^2$

Date ....

→ MATRIX FORM OF A LINEAR TRANSFORMATION

If $T: U \to V$ be a linear transformation then there exists a unique matrix such that
$T(X) = A \cdot X$
For every vector $X \in U$. This matrix A is called matrix representation of T

NOTE
If dimension of $U = m$ & V has dimension n then matrix representation of T i.e. A is of order $n \times m$

→ Procedure to write A
Step 1: Find the basis for the vector space U (if not given we always consider standard basis)
Step 2: Find the image of the basis elements of U under linear transformation T
Step 3: Write the image elements in terms of linear combination of the basis elements of V
Step 4: Write the corresponding coefficients in the columns of a matrix

Date ....

Q $T: R^2 \to R^2$

$T(x, y) = (x+y, x-y)$

Basis of $R^2 = \{(1,0), (0,1)\} = B_1$

$T(1,0) = (1, 1)$; $T(0,1) = (1, -1)$

$(1,1) = a(1,0) + b(0,1)$

$(1,1) = (a, b)$

$a = b = 1$

$(1,-1) = a(1,0) + b(0,1)$

$(1,-1) = (a, b)$

$a = 1 \quad b = -1$

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Q $T: R^2 \to R^2$

$T(x, y) = (x+y, x-y)$

$B(u) = \{(1,1), (1,0)\}$

$T(1,1) = (2, 0)$ $\quad T(1,0) = (1,1)$

$(2,0) = a(1,0) + b(0,1)$

$(2,0) = (a, b)$

$a = 2 \quad b = 0$

$(1,1) = (a, b)$

$a = b = 1$

$$\begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

Date ....

Q Find matrix representation of the linear transformation

$T: R^3 \to R^3$ such that

$T(x_1, x_2, x_3) = (x_1 - x_2 + x_3,\ 2x_1 + 3x_2 - \frac{1}{2}x_3,$

$x_1 + x_2 - 2x_3)$

where $B(u) = (1,0,0)\ (0,1,0)\ (0,0,1)$

$B(V) = (1,1,0)\ (1,2,3),\ (-1,0,1)$

$T(1\ 0\ 0) = (1, 2, 1)$

$T(0\ 1\ 0) = (-1, 3, 1)$

$T(0\ 0\ 1) = (1, -1/2, -2)$

$(1, 2, 1) = a(1,1,0) + b(1,2,3) + c(-1,0,1)$

$1, 2, 1 = (a+b-c,\ a+2b,\ 3b+c)$

$a+b-c = 1$ $\quad a+2b = 2$ $\quad 3b+c = 1$

$a = 2-2b$ $\quad c = 1-3b$

$2-2b+b-1+3b = 1$

$1+2b = 1$

$b = 0$ $\quad a = 2$ $\quad c = 1$

$-1, 3, 1 =$

$a+2b = 3$ $\quad 3b+c = 1$

$a+b-c = -1$ $\quad a = 3-2b$ $\quad c = 1-3b$

$3-2b+b-1+3b = -1$ $\quad a = 3+3$ $\quad c = 1+\frac{9}{2}$

$2+2b = -1$ $\quad a = 6$

$2b = -3$ $\quad c = \frac{11}{2}$

$b = -3/2$

Spiral

Date ....

$$\left(1, -\frac{1}{2}, -2\right) = (a+b-c,\ a+2b,\ 3b+c)$$

$$a = -\frac{1}{2} - 2b$$

$$c = -2 - 3b$$

$$1 = -\frac{1}{2} - 2b + b + 2 + 3b$$

$$1 = \frac{3}{2} + 2b$$

$$-\frac{1}{2} = 2b$$

$$b = -\frac{1}{4}$$

$$a = -\frac{1}{2} - \cancel{2}\left(\frac{-1}{\cancel{2}4}\right)$$

$$a = 0$$

$$c = -2 + \frac{3}{4}$$

$$c = -\frac{5}{4}$$

$$\begin{bmatrix} 2 & 6 & 0 \\ 0 & -3/2 & -1/4 \\ 1 & 11/2 & -5/4 \end{bmatrix}$$

Rank = 3

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Date ....

→ Rank - nullity theorem

If $T : U \rightarrow V$ be a linear transformation then

$$\text{Rank}(T) + \text{Nullity}(T) = \dim(U)$$

→ One - One Linear Transformation

A linear transformation $T : U \rightarrow V$ is said to be one - one if $\forall\ u_1, u_2 \in U$ we have $T(u_1) = T(u_2) \Rightarrow u_1 = u_2$

* For every one - one L.T. nullity = 0 also Ker (T) only has a element which is a zero element.

Date ....

Rank - Nullity Theorem

If $V(F)$ & $W(F)$ are 2 vector spaces and $T: V \rightarrow W$ is a linear tranformation then

$$Nullity(T) + Rank(T) = Dim(V)$$

As $dim(V) = n$

$\exists B = \{v_1, v_2 \dots v_p, v_{p+1}, v_{p+2} \dots v_n\}$ is the basis of V

since $Ker(T)$ is the subspace of finite dim V. so $Ker(T)$ also has a finite dimension

Let $dim(Ker(T)) = p$
i.e. $nullity(T) = p$

$\exists B_1 = \{v_1, v_2 \dots v_p\}$ is basis of $Ker(T)$

also $(T(v_1) = T(v_2) = \dots = T(v_p) = 0$

Let
$B_2 = \{T(v_{p+1}), T(v_{p+2}) \dots T(n)\}$

Now check linear independency

$$\sum_{i=p+1}^{n} \alpha_i T(v_i) = 0$$

as $T \rightarrow$ linear

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$$\sum_{i=p+1}^{n} T(\alpha_i v_i) = 0$$

$$\Rightarrow T(\alpha_{p+1} v_{p+1}) + T(\alpha_{p+2} v_{p+2}) \cdots\cdots T(\alpha_n v_n) = 0$$

$$\Rightarrow T\left[\sum_{i=p+1}^{n} \alpha_i v_i\right] = 0$$

$$\Rightarrow \sum_{i=p+1}^{n} \alpha_i v_i \in \ker(T)$$

Since B_1 is basis of $\ker(T)$

$$\Rightarrow \sum_{i=p+1}^{n} \alpha_i v_i = \sum_{i=1}^{p} \beta_i v_i$$

$$\Rightarrow \beta_1 v_1 + \beta_2 v_2 \cdots \beta_p v_p - \alpha_{p+1} v_{p+1} - \alpha_{p+2} v_{p+2} \cdots \alpha_n v_n = 0$$

Since

B is the basis of V

$\beta_1 = \beta_2 \cdots \beta_p = 0 \quad \Rightarrow \alpha_{p+1} = \alpha_{p+2} \cdots \alpha_n = 0$

Hence B_2 is L.I.

Now,

let $y \in R(T)$ be any element

$\therefore \exists x \in V$ such that $y = T(x)$

Date ....

since $x \in V$ & $B \to$ basis of V

$$\Rightarrow x = \sum_{i=1}^{n} \alpha_i v_i$$

$$\Rightarrow T(x) = T\left(\sum_{i=1}^{n} \alpha_i v_i\right) = \alpha_i T\left(\sum_{i=1}^{n} \cancel{\alpha_i} v_i\right)$$

$$\Rightarrow T(x) = \sum_{i=1}^{p} \alpha_i T(v_i) + \sum_{i=p+1}^{n} \alpha_i T(v_i)$$

Bcoz $T(v_i) = 0$ for $i = 1, 2, 3 \ldots p$

$$T(x) = \sum_{i=p+1}^{n} \alpha_i T(v_i)$$

$\Rightarrow y = T(x)$ is linear comb. of elements of B_2
$\Rightarrow B_2 \to$ span $R(T)$

Hence B_2 forms Basis of $R(T)$

$$\Rightarrow \dim(B_2) = (n-p)$$

$$\Rightarrow \text{Ran}(T) = n - \text{nullity}(T)$$

$$\Rightarrow n = \text{Rank}(T) + \text{nullity}(T)$$